

Haoran Sun

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Research Profile

My research combines game-theoretic modelling of industrial policy and taxation with applied microeconomic analysis of firm productivity, innovation support, and technology adoption. On the theoretical side, I develop models of industrial policy under network externalities and of international taxation of digital businesses. On the empirical side, I use ONS microdata to analyse high-growth business dynamics and identify the firm-level effects of public policy. My published work includes one ABS 4 and two ABS 3 journal articles, alongside two papers under review and three further papers at advanced stages. My forward research programme links theoretical predictions on network externalities to empirical analysis of AI adoption and public procurement, and will underpin future external funding applications, including an ESRC New Investigator Grant.

Job Market Paper

Industrial Policy with Network Externalities: Race to the Bottom vs. Win-Win Outcome

Nigar Hashimzade and Haoran Sun

Studies industrial policy in a two-country model with strategic trade and R&D under network externalities. Shows that policy competition can generate either a race to the bottom or a win-win welfare outcome, depending on the strength of network effects, the degree of product differentiation, and whether innovation is product- or process-based.

Employment

- 2025– Researcher (PDRA), Durham University Business School
Smart & Scale Up project with an external industry partner
Firm-level analysis of high-growth business dynamics in North East England using BSD, LSBS, and econometric methods (decomposition analysis, treatment effect estimation).
Research report delivered to partner; findings informed advisory targeting and programme design.
- 2023–2025 Research Fellow, University of St Andrews
ESRC Funded (Award ES/W010259/1)
Data management, econometric analysis, and co-authorship across work packages examining SME finance, investment, and productivity using firm-level microdata.

Education

- 2017–2023 PhD in Economics, Durham University
ESRC Funded – Tax Incentives for Innovation
Thesis: [The Economics of Innovation, Investment, and Taxation](#)
- 2018 Visiting PhD in Economics, University of Exeter
- 2015–2017 MSc in Economics (Distinction), Durham University
- 2012–2015 BA in Economics (2:1 Honours), Durham University

Publications

Spatial Myopia, Financial Hubs and the “Golden Triangle”: Tracing the Uneven Landscape of Exogenous Equity Investments in the UK

Ross Brown, Haoran Sun, Augusto Rocha, Marc Cowling
British Journal of Management (ABS 4)

Unpacking the Money-Distance Nexus: Mapping the Spatial Configuration of Exogenous Entrepreneurial Finance across UK Entrepreneurial Ecosystems

Ross Brown, Augusto Rocha, Haoran Sun, Marc Cowling
Small Business Economics (ABS 3)

Why do Owls Fly While Ostriches Flounder? SME Advice, Impacts and Entrepreneurial Outcomes

Ross Brown, Marc Cowling, Weixi Liu, Sabrina Song, Haoran Sun
Journal of Small Business Management (ABS 3)

Papers Under Review

Can We See it in the Numbers Yet? Artificial Intelligence, Firm-Level Productivity and SMEs

Ross Brown, Haoran Sun, Marc Cowling, David Dowell
Journal of Technology Transfer (ABS 3), under review

Does “Slow” Thinking “Speed Up” Performance? The Impact of “Guided Preparation” on Innovation and Productivity in UK SMEs

Ross Brown, Haoran Sun, Weixi Liu, Marc Cowling, Sabrina Song
Journal of Technology Transfer (ABS 3), under review

Working Papers

The Effect of Direct and Indirect Support for Innovation on Productivity: Evidence from Young UK Businesses

John Moffat, Haoran Sun^{*}
Main analysis complete; further empirical work in progress

Taxing the Clouds: An Economic Analysis

Nigar Hashimzade, Haoran Sun^{*}
Manuscript being prepared for submission

Work in Progress

Finance, Growth and Productivity in Family-Owned SMEs

Ross Brown, Marc Cowling, Haoran Sun
ERC-DBT grant awarded (2026); analysis of LSBS data in progress

Do Science Parks Act as a Magnet for Investment? The Role of University Proximity

Ross Brown, Marc Cowling, Haoran Sun
Dataset constructed; empirical analysis forthcoming

Grants

- 2026 ERC–DBT Small Grant Funding: A Study of Finance, Growth and Productivity in Family-Owned SMEs, £5,000
Ross Brown, Marc Cowling, Haoran Sun (Co-Investigator)
- 2024 ERC–DBT Small Grant Funding: The Impact of “Guided Preparation” on Innovation and Productivity in UK SMEs, £5,000
Ross Brown, Marc Cowling, Haoran Sun (Co-Investigator)

Policy & Stakeholder Engagement

- 2025 Presentation of North East High Growth Business report to a major North East business support provider
Findings directly informed advisory targeting and programme design, Durham
- 2025 Smart & Scale Up project kickoff meeting with a major North East business support provider
Stakeholder partnership briefing, Durham University Business School
- 2025 ERC–DBT Longitudinal Small Business Survey Dissemination Event
Policy dissemination to government and industry audience, London

Academic Presentations

- 2024 ZEW: Competition and Innovation Workshop, Montenegro
- 2024 Research Review, ESRC Team Project Meeting (Award ES/W010259/1), Oxford Brookes University
- 2024 “Unpacking the Money-Distance Nexus,” Internal Research Seminar, University of St Andrews

Teaching

- 2026 Guest Lecturer – *What Shaped This Data? Platform Economics and the Limits of Analytics*
MSc Digital Marketing and MSc Digital Marketing Strategy, Newcastle University
- 2020 Teaching Assistant – Macroeconomics (UG), Durham University
Led seminar groups of up to 20 students; facilitated presentations, debates, and problem sets; marked coursework and provided individual feedback
- 2019 Teaching Assistant – Corporate Finance (UG), Durham University
Marked coursework and provided individual feedback
Teaching competencies: Microeconomics, Macroeconomics, Econometrics, Microeconometrics, Public Economics (UG and PG).

Honours & Professional Activities

- 2017–2022 NINE Doctoral Training Partnership Studentship, ESRC
- 2017–2021 Research Training Support Grant, ESRC

Technical Skills & Languages

Software: Stata, Python, R, $\LaTeX$, Maple, Matlab, EViews, SPSS, NVivo

English: Professional proficiency

Mandarin: Native proficiency

References

References available on request.